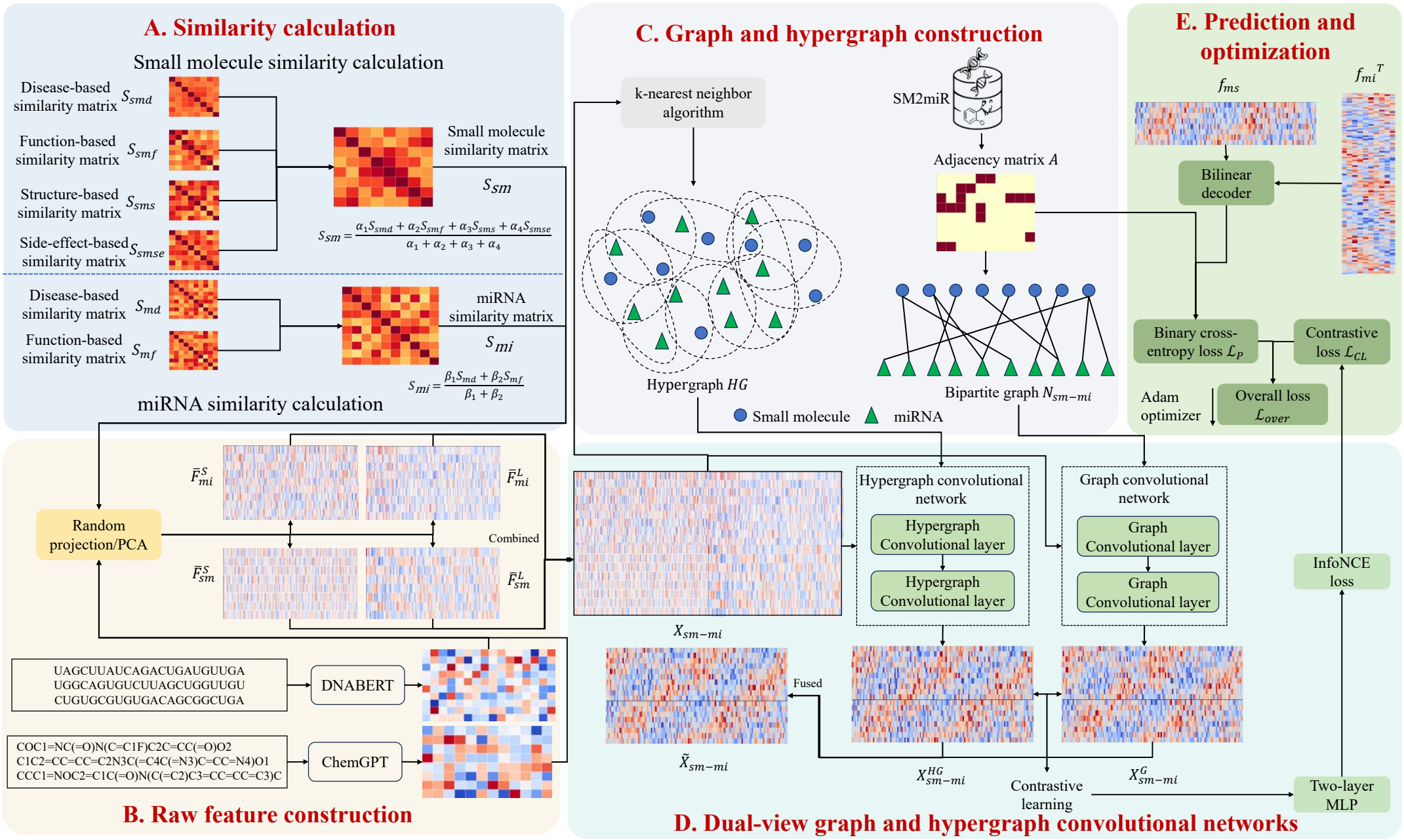

A. Similarity calculation
Small molecule similarity calculation
Disease-based similarity matrix S_{smd}
Function-based similarity matrix S_{smf}
Structure-based similarity matrix S_{sms}
Side-effect-based similarity matrix S_{smse}
Small molecule similarity matrix
S_{sm}
$S_{sm} = \frac{\alpha_1 S_{smd} + \alpha_2 S_{smf} + \alpha_3 S_{sms} + \alpha_4 S_{smse}}{\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4}$
Disease-based similarity matrix S_{md}
Function-based similarity matrix S_{mf}
miRNA similarity matrix
S_{mi}
$S_{mi} = \frac{\beta_1 S_{md} + \beta_2 S_{mf}}{\beta_1 + \beta_2}$
miRNA similarity calculation
C. Graph and hypergraph construction
k-nearest neighbor algorithm
SM2miR
Adjacency matrix A
Hypergraph HG
Bipartite graph N_{sm-mi}
Small molecule
miRNA
E. Prediction and optimization
f_{ms}
f_{mi}^T
Bilinear decoder
Binary cross-entropy loss $\mathcal{L}_P$
Contrastive loss $\mathcal{L}_{CL}$
Adam optimizer
Overall loss $\mathcal{L}_{over}$
$\bar{F}_{mi}^S$
$\bar{F}_{mi}^L$
Random projection/PCA
Combined
$\bar{F}_{sm}^S$
$\bar{F}_{sm}^L$
UAGCUUAUCAGACUGAUGUUGA
UGGCAGUGUCUUAGCUGGUUGU
CUGUGCGUGUGACAGCGGCUGA
DNABERT
COC1=NC(=O)N(C=C1F)C2C=CC(=O)O2
C1C2=CC=CC=C2N3C(=C4C(=N3)C=CC=N4)O1
CCC1=NOC2=C1C(=O)N(C(=C2)C3=CC=CC=C3)C
ChemGPT
B. Raw feature construction
X_{sm-mi}
Hypergraph convolutional network
Hypergraph Convolutional layer
Hypergraph Convolutional layer
Graph convolutional network
Graph Convolutional layer
Graph Convolutional layer
InfoNCE loss
Fused
$\tilde{X}_{sm-mi}$
X_{sm-mi}^{HG}
Contrastive learning
X_{sm-mi}^{G}
Two-layer MLP
D. Dual-view graph and hypergraph convolutional networks

